

Golli protein negatively regulates store depletion-induced calcium influx in T cells.

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Evidence abstract

Calcium influx is crucial for T cell activation and differentiation.

The detailed regulation of this process remains unclear.

We report here that golli protein, an alternatively spliced product of the myelin basic protein gene, plays a critical role in regulating calcium influx in T cells.

Golli-deficient T cells were hyperproliferative and showed enhanced calcium entry upon T cell receptor stimulation.

We further found that golli regulates calcium influx in T cells through the inhibition of the store depletion-induced calcium influx.

Mutation of the myristoylation site on golli disrupted its association with the plasma membrane and reversed its inhibitory action on Ca^{2+} influx, indicating that membrane association of golli was essential for its inhibitory action.

These results indicate that golli functions in a unique way to regulate T cell activation through a mechanism involving the modulation of the calcium homeostasis.

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